AgriCrop: Geospatial Plant Disease & Soil Moisture Intelligence Network

1. Project Introduction

AgriCrop is an advanced AI-driven initiative designed to modernize precision agriculture. By leveraging geospatial data and real-time sensor analytics, the system provides an intelligent network capable of monitoring crop health and optimizing soil moisture management.

2. Project Objectives

- **Geospatial Analysis:** To utilize satellite and aerial imagery for large-scale crop health monitoring.
- **Disease Detection:** To implement computer vision models for the early identification of plant diseases.
- **Soil Moisture Intelligence:** To analyze soil data and provide actionable insights for efficient irrigation, reducing water wastage.

3. Technical Architecture

Component	Technology Used
AI/ML Framework	Python, TensorFlow, OpenCV
Data Processing	Geospatial analysis libraries (GDAL/Rasterio)
Backend Integration	API-driven architecture for real-time monitoring

4. Implementation Strategy

1. **Data Acquisition:** Collecting agricultural data and sensor inputs.
2. **Model Training:** Training CNN models for disease detection.
3. **Network Intelligence:** Deploying an integrated network to map moisture levels and crop status.

5. Conclusion

The AgriCrop project addresses critical challenges in modern agriculture by integrating AI into field operations, fostering sustainable farming practices through data-backed intelligence.